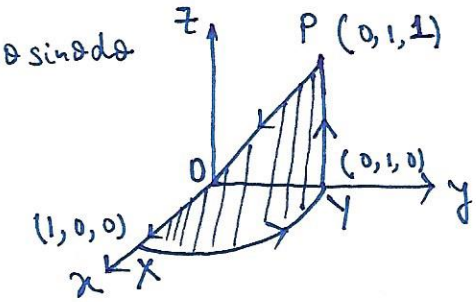
1. Verify Stokes theorem for a vector function $\vec{v}$ expressed in spherical co-ordinates (r, θ, φ) as:

$$\vec{v} = (r\cos^2\theta)\,\hat{r} - (r\cos\theta\sin\theta)\,\hat{\theta} + 3r\,\hat{\varphi}$$ in the shaded region below:

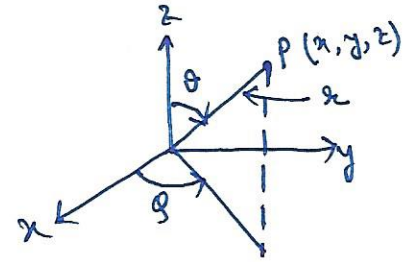
$$\left[\vec{v}\cdot\vec{dl} = r\cos^2\theta\, dr - r^2\cos\theta\sin\theta\, d\theta + 3r^2\sin\theta\, d\varphi\right]$$



<u>Along x-axis:</u>

$\theta = \pi/2$, $\varphi = 0$, $r: 0 \to 1$

Along xy plane θ is always $\pi/2$
(θ measured w.r.t z-axis)



$\vec{v} = 3r\,\hat{\varphi}$ and $\vec{dl} = dl_r\,\hat{r} + dl_\theta\,\hat{\theta} + dl_\varphi\,\hat{\varphi}$

$$= dr(\hat{r}) + (r\,d\theta)\,\hat{\theta} + (r\sin\theta\, d\varphi)\,\hat{\varphi}$$

$$\vec{v}\cdot\vec{dl} = 3r^2\sin\theta\, d\varphi = 3r^2\, d\varphi. \quad (\theta = \pi/2)$$

$\int_{OX} \vec{v}\cdot\vec{dl} = \int 3r^2\, d\varphi$. But $\varphi = 0$ along OX and thus $\int \vec{v}\cdot\vec{dl} = 0$ along OX

<u>Along XY</u>

$r = 1$, $dr = 0$. $\theta = \pi/2$, $d\theta = 0$, φ: 0 to $\pi/2$

$$\vec{v}\cdot\vec{dl} = 3r\cdot r\sin\theta\, d\varphi = 3r^2\, d\varphi \quad (\theta = \pi/2)$$

$$\int_{XY} \vec{v}\cdot\vec{dl} = 3r^2\int_0^{\pi/2} d\varphi = 3r^2\,\frac{\pi}{2} = \frac{3\pi}{2} \quad (r = 1)$$

<u>Along YP</u>

In yz plane $\varphi = \pi/2$, $d\varphi = 0$; $\vec{v}\cdot\vec{dl} = (r\cos^2\theta\, dr - r^2\cos\theta\sin\theta\, d\theta)$

Here both r and θ are variables, we would have to convert $\vec{v}\cdot\vec{dl}$ as function of one variable, say θ.

In spherical co-ordinates $y = r\sin\theta\sin\varphi = r\sin\theta$ $[\varphi = \pi/2]$
$= 1$ $[\vec{OY} = \hat{y}]$

Thus $y = r\sin\theta = 1$

Since along PO, $y = z$ is the eqn. of the line, the angle which PO makes with z-axis (θ) is $\pi/4$ and along OY $\theta = \pi/2$. Thus YP, θ should vary from $\frac{\pi}{2}$ to $\frac{\pi}{4}$.

$$\vec{v} \cdot d\vec{l} = r\cos^2\theta \, dr - r^2 \cos\theta \sin\theta \, d\theta$$

Convert $r \to \theta$

$$y = r\sin\theta = 1 \Rightarrow r = \frac{1}{\sin\theta} = \operatorname{cosec}\theta$$

$$dr = -\operatorname{cosec}\theta \cot\theta \, d\theta = -\frac{\cos\theta}{\sin^2\theta} d\theta$$

$$\vec{v} \cdot d\vec{l} = -r\cos^2\theta \frac{\cos\theta}{\sin^2\theta} d\theta - r^2 \cos\theta \sin\theta \, d\theta$$

$$= \frac{-1}{\sin^3\theta} \cos^3\theta \, d\theta - \frac{\cos\theta \sin\theta}{\sin^2\theta} d\theta$$

$$= -\frac{\cos\theta}{\sin^3\theta}\left[\cos^2\theta + \sin^2\theta\right] d\theta = -\frac{\cos\theta}{\sin^3\theta} d\theta$$

$$\int_{YP} \vec{v} \cdot d\vec{l} = \int_{\pi/2}^{\pi/4} \left[-\frac{\cos\theta}{\sin^3\theta} d\theta\right] = -\int_{\pi/2}^{\pi/4} \frac{1}{\sin^3\theta} d(\sin\theta) = \frac{1}{2\sin^2\theta}\Big|_{\pi/2}^{\pi/4} = \frac{1}{2}$$

<u>Along PO</u>

$\varphi = \pi/2$ (yz plane), $\theta = \pi/4$, $d\varphi = d\theta = 0$, $r: \sqrt{2} \to 0$

$$\int_{PO} \vec{v} \cdot d\vec{l} = \int_{r=\sqrt{2}}^{0} r\cos^2\theta \, dr = \left(\frac{1}{\sqrt{2}}\right)^2 \frac{r^2}{2}\Big|_{\sqrt{2}}^{0} = -\frac{1}{2}$$

$$\boxed{\oint \vec{v} \cdot d\vec{l} = 0 + \frac{3\pi}{2} + \frac{1}{2} - \frac{1}{2} = \frac{3\pi}{2}}$$

Now we have to calculate the surface integral of $(\vec{\nabla} \times \vec{v})$ on two surfaces - one surface along yz plane (boundary of surface running anticlockwise) and another along xy plane (boundary of surface running anticlockwise).

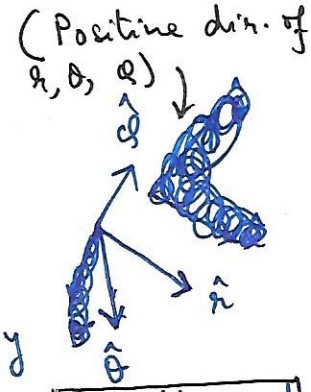
To find $\vec{\nabla}\times\vec{v}$, we have to apply the formula for spherical co-ordinates

$$\vec{\nabla}\times\vec{v} = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta\cdot\underbrace{3r}_{v_\varphi}) - \frac{\partial}{\partial\varphi}(\overbrace{-r\sin\theta\cos\theta}^{v_\theta})\right]\hat{r}$$

$$+\frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}(\underbrace{r\cos^2\varphi}_{v_r}) - \frac{\partial}{\partial r}(\underbrace{3r}_{v_\varphi}\cdot r)\right]\hat{\theta}$$

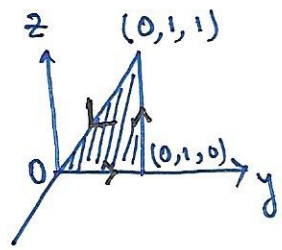
$$+\frac{1}{r}\left[\frac{\partial}{\partial r}(r\cdot\underbrace{r\sin\theta\cos\varphi}_{v_\theta}) - \frac{\partial}{\partial\theta}(\underbrace{r\cos^2\theta}_{v_r})\right]\hat{\varphi}$$

$$= (3\cot\theta\,\hat{r} - 6\,\hat{\theta})$$



(in spherical polar)

<u>Surface in yz plane</u>



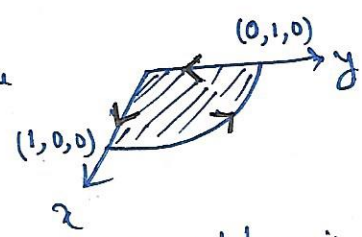
An area element can be expressed in spherical polar co-ordinates as

$\vec{da} = -dh_r\, dl_\theta\, \hat{\varphi}$ [see Lecture slides and Griffiths pg 40]

$= -dr\, r\, d\theta\, \hat{\varphi}$ [minus sign appears because of the sense of traverse along the boundary of surface (anticlockwise) corresponds to the opposite direction of φ (out of page) as compared to the usual direction of φ (into the page) in spherical co-ordinates]

Thus $(\vec{\nabla}\times\vec{v})\cdot\vec{da} = 0$

<u>Surface in x-y plane</u>



Similarly surface in x-y plane ($\theta = \text{const} = \pi/2$) can be expressed as

$\vec{da} = -dl_r\, dl_\varphi\, \hat{\theta}$ [minus sign appearing due to reasons similar to above - <u>sense of traverse</u> along the boundary contributing to the sign]

$= -dr\, r\sin\theta\, d\varphi\, \hat{\theta}$

$= -r\, dr\, d\varphi\, (\hat{\theta})$ $[\theta = \pi/2]$

Thus $\vec{\nabla}\times\vec{v}\cdot\vec{da} = +6\, r\,dr\,d\varphi$

$$\iint \vec{\nabla}\times\vec{v}\cdot da = +6\int_{r=0}^{1} r\,dr \int_{\varphi=0}^{\pi/2} d\varphi$$

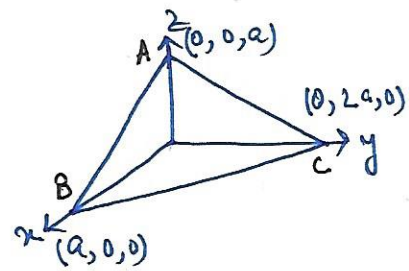
Surface represents a circle of radius $(r=1)$

$$= +\frac{6}{2}\cdot\frac{\pi}{2} = +\frac{3\pi}{2}$$

$$\boxed{\iint_{\substack{\text{entire}\\\text{surface}}} \vec{\nabla}\times\vec{v}\cdot\vec{da} = \frac{3\pi}{2} = \oint \vec{v}\cdot\vec{dl}}$$

(Stoke's theorem verified)

2. Check Stokes theorem for the function $\vec{v} = y\hat{z}$ using the triangular surface as shown.



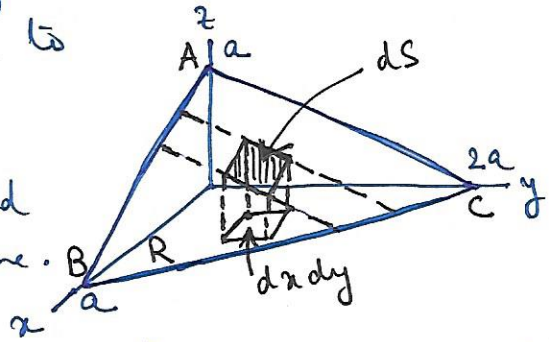
We have to show $\iint_S (\vec{\nabla}\times\vec{v})\cdot\hat{n}\,dS$

$$= \oint \vec{v}\cdot\vec{dl}$$

S is the surface of the triangle and the closed path is the outer boundary of the surface. Let us calculate the surface integral first.

We will use the projection method to calculate the surface integral.

Choose an elementary area dS and take its projection on the x-y plane.



$$\iint (\vec{\nabla}\times\vec{v})\cdot\hat{n}\,dS = \iint_R (\vec{\nabla}\times\vec{v})\cdot\hat{n}\,\frac{dx\,dy}{|\hat{n}\cdot\hat{k}|} \quad \text{[R is the region in x-y plane]} \quad \text{——— (a)}$$

$\hat{n}$ is unit normal to dS - eqn of the plane is $\frac{x}{a}+\frac{y}{2a}+\frac{z}{a}=1$

or $2x+y+2z=2a$ ——— (b)

$\vec{n} = \vec{\nabla}(2x+y+2z) = 2\hat{i} + \hat{j} + 2\hat{k}$

$\hat{n} = \dfrac{2\hat{i}+\hat{j}+2\hat{k}}{3}$ $\quad \therefore \hat{n}\cdot\hat{k} = \dfrac{2}{3}$

$$\boxed{(\vec{\nabla}\times\vec{v})\cdot\hat{n} = \hat{i}\cdot\left(\frac{2}{3}\hat{i}+\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}\right) = \frac{2}{3}}$$

It is clear $\vec{\nabla}\times\vec{v} = \hat{i}$. Eqn. (a) reduces to:

$$\iint (\vec{\nabla}\times\vec{v})\cdot\hat{n}\, dS = \iint (\vec{\nabla}\times\vec{v})\cdot\hat{n}\, \frac{dx\,dy}{|\hat{n}\cdot\hat{k}|}$$

$$= \frac{2}{3}\times\frac{3}{2}\iint dx\,dy = \iint dx\,dy$$

<u>Along BC</u>

$2x+y=2a$ is the eqn of line. Thus we can choose the limit of y from 0 to $(2a-x)$ and then run x from 0 to a.

$$\iint dx\,dy = \int_0^{2(a-x)} dy \int_{x=0}^{a} dx \quad \text{[y integral should be computed first]}$$

$$= \int_{x=0}^{a} 2(a-x)\,dx = 2a\cdot a - \frac{1}{2}\cdot 2x^2\Big|_0^a = 2a^2 - a^2 = a^2$$

Thus $\boxed{\iint (\vec{\nabla}\times\vec{v})\cdot\vec{dS} = a^2}$

Let's compute the line integral now for different segments.

$\vec{v}\cdot\vec{dl} = v_x\,dx + v_y\,dy + v_z\,dz = v_z\,dz = y\,dz$

$\int_{AB} \vec{v}\cdot\vec{dl} = 0$ [$\because y=0$ along AB]

$\int_{BC} \vec{v}\cdot\vec{dl} = \int y\,dz$ [along BC, $z=0$, so $dz=0$]
$= 0$

$\int_{AC} \vec{v}\cdot\vec{dl} = \int y\,dz$. Along AC, $y+2z=2a$, i.e $y = 2(a-z)$

$= \int_{z=0}^{a} 2(a-z)\,dz = a^2$

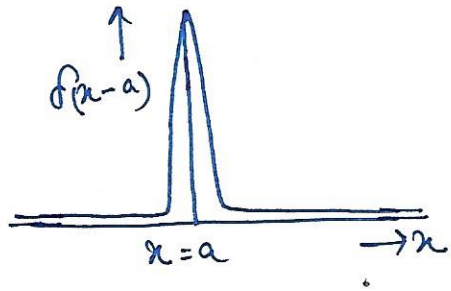
$$\therefore \boxed{\oint \vec{v}\cdot\vec{dl} = a^2 = \iint (\vec{\nabla}\times\vec{v})\cdot\vec{dS}}$$

3. (a) $\displaystyle\int_{-\infty}^{+\infty} e^{-x}\,\delta(x-\ln 2)\,dx$

We know $\displaystyle\int_{x_1}^{x_2} f(x)\,\delta(x-a)\,dx = f(a)\int_{x_1}^{x_2}\delta(x-a)\,dx$

$= f(a)$, if $a \in (x_1, x_2)$

$= 0$ otherwise



By definition $\displaystyle\int_{x_1}^{x_2}\delta(x-a)\,dx = 1$ if the range of integration covers the location of the spike $x=a$.

$$\int_{-\infty}^{\infty} e^{-x}\,\delta(x-\ln 2)\,dx = e^{-\ln 2}\int_{-\infty}^{+\infty}\delta(x-\ln 2)\,dx$$

$$= e^{-\ln 2} = \frac{1}{e^{\ln 2}} = \frac{1}{2}$$

(b) $\displaystyle\int_{1}^{3} x^3\,\delta(x-\pi)\,dx$

$= \pi^3\displaystyle\int_{1}^{3}\delta(x-\pi)\,dx = \pi^3 \times 0$ [$\because$ domain of integration does not lie with $x=\pi$]

(c) $\displaystyle\int_{-1/2}^{7/2} \sin x\,\delta(x-\pi)\,dx$

$= \sin\pi\displaystyle\int_{-1/2}^{7/2}\delta(x-\pi)\,dx = \sin\pi \times 1 = 0$

[$x=\pi$ is within domain of integration hence its integral is unity]